

P3↔P2 Evolution Mechanisms Inheritance

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This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026). It does not introduce new architectural commitments. Its sole contribution is to formalize, in derivation form, the inheritance edge by which Paper 3’s four-locus evolution feed extends Paper 2’s three evolution mechanisms to inter-Self scope — adding new sources to existing mechanisms rather than creating new mechanisms.

Abstract

Paper 2 establishes three evolution mechanisms through which a Self evolves its coordination capabilities: instinct evolution (undirected mutation through LLM and infrastructure upgrades), DNA evolution (directed selection through governance-authorized substrate changes), and action-feedback evolution (empirical loop from action-layer experience back into governed DNA refinement). These three mechanisms operate within a single Self’s home governance architecture. Paper 3 introduces the four-locus evolution feed, which extends the architecture to inter-Self scope by creating pathways through which Full Aspect Integration (FAI) event outcomes enter each participating Self’s existing evolution mechanisms at dissolution. This note formalizes the inheritance edge: Paper 3 adds *sources*, not *mechanisms*. The four loci are input channels into Paper 2’s established machinery, not a parallel or replacement evolutionary architecture. Two loci are active pathways (action-feedback and directed selection receive FAI-derived content), one locus is an explicit protection (instinct evolution is out of scope for inter-Self exchange), and one locus is the routing mechanism that governs which pathway receives which content at dissolution. Any adversarial claim about a novel inter-organizational AI learning architecture must address that FAI learning enters through Paper 2’s action-feedback and directed selection mechanisms, both established prior art.

1. Why the inheritance must be made explicit

The extension of an existing mechanism to a broader scope is not automatically recognized as inheritance. Without explicit formalization, the four-locus evolution feed is susceptible to two misreadings. The first misreading is under-claim: treating the four loci as mere bookkeeping around Paper 2’s §7, with no architectural content beyond noting that FAI events exist. The second misreading is over-claim: treating the four loci as a new evolutionary architecture that competes with

or replaces Paper 2's three mechanisms, and therefore as requiring independent prior-art defense.

Both misreadings are wrong, and the architectural ground between them is where this inheritance note operates. Paper 3's Claim 4 introduces four loci of fresh architectural specification — the hand-off mechanism at dissolution, the per-mechanism feed structure, the layer-routing rule, and the asymmetric ingestion property — while leaving Paper 2's three evolution mechanisms entirely unchanged. The mechanisms operate at home perimeter, under home governance, identically to how they operated before any FAI event. What changes is the set of available inputs. This is the inheritance claim: same mechanisms, new sources. Formalizing it closes the adversarial gap on both sides simultaneously.

2. Source commitment at Paper 2 scope: the three evolution mechanisms

Paper 2 §7 commits to three evolution mechanisms operating in productive tension within every Self.

Instinct evolution is undirected mutation. LLM upgrades and substrate-platform infrastructure upgrades sharpen what the Self can do at the instinct and infrastructure layer without modifying DNA content. These upgrades arrive from upstream providers that the Self does not control; they are mutation-like in that capability changes may be good or bad in any specific case and are not goal-directed from the Self's perspective. Governance integration operates through verification substrates, selective retirement of superseded reasoning, and orchestration rules that pin high-stakes decisions to the reasoning layer regardless of instinct capability. The key property for this inheritance note is independence: instinct evolution operates without governance authorization of the upgrade itself, and without any input from other Selves.

DNA evolution is directed selection. Governance-authorized changes to the orchestration substrate are evaluated and accepted under selection criteria that are themselves substrate content — human-governed, inspectable, and modifiable. The directionality is external and persistent: humans define success criteria and hold ongoing authority over acceptance decisions. LLMs may draft proposed changes under human direction; humans hold authority over the acceptance decisions. This is the primary governance-controlled evolution mechanism within a Self.

Action-feedback evolution closes the empirical loop. Operational records accumulated in the action layer generate evidence — what worked, what failed, what surprised — and that evidence feeds into governed proposals for DNA refinement. The loop is discrete and auditable: humans or LLMs operating under human direction propose substrate changes from action patterns; humans authorize the changes through the proposal-and-acceptance machinery. The loop exists, it is governed, and it cannot silently drift the DNA without human authorization. This mechanism is the only one through which a Self learns from its own operational experience rather than from external upgrades or human design decisions.

At intra-Self scope, all three mechanisms operate within one Self's home governance architecture, applying concurrently at cell, aspect, and Self levels, with different humans governing different

levels under appropriate authority scopes.

3. Extension to inter-Self scope: the four-locus evolution feed

Paper 3 does not create new evolution mechanisms. It creates four loci — two active pathways, one protective boundary, and one routing mechanism — through which FAI-origin content reaches Paper 2's existing mechanisms at each participating Self's home perimeter.

Locus 1 — Action-feedback pathway (D1.17). When a FAI event dissolves, event outcomes and operational records enter each participating Self's home action layer. Once there, the home action-feedback mechanism processes this content identically to content generated within that Self's own operations. The action layer accumulates it; the proposal-and-acceptance machinery can generate DNA refinement proposals from it; humans authorize those proposals under home authority. FAI learning enters through the established action-feedback pathway. No new mechanism is introduced; the source set is expanded.

Locus 2 — Directed selection pathway (D1.18). FAI-origin DNA content is made available for each participating Self's directed selection consideration. Home governance reviews it through the absorption protocol and, if authorized, absorbs it through a directed selection event — the same governance-controlled DNA evolution mechanism Paper 2 establishes at intra-Self scope. Selection criteria remain home substrate content; absorption decisions remain under home authority. FAI learning enters through the established directed selection pathway. No new mechanism is introduced; the source set is expanded.

Locus 3 — Instinct non-crossing (D1.19). This locus is not an active pathway. It is the explicit commitment that LLM weights and instinct-layer content do not cross the inter-Self perimeter during FAI events. Instinct evolution stays entirely home — LLM upgrades within each Self are independent of FAI participation; no inter-Self influence on mutation exists. Locus 3 is the protection of the instinct layer from inter-Self influence, not an evolution pathway. It corresponds to the instinct/reasoning separation Paper 2 establishes at intra-Self scope and Paper 3 carries through at inter-Self scope without modification. The practical implication is that instinct evolution — the one mechanism that operates without governance authorization of the change itself — cannot be triggered or influenced by FAI events. The only mechanisms that receive FAI inputs are the two that operate under governance authorization.

Locus 4 — Hand-off boundary (D1.20). The governance-specified routing mechanism at dissolution. When the FAI shared substrate dissolves, the content it contains is routed to the appropriate home evolution loci per the layer it belongs to: DNA-layer content routes to directed selection (Locus 2); action-layer content routes to action-feedback (Locus 1); instinct-layer content is not in scope (Locus 3). The hand-off boundary is not an evolution mechanism itself; it is the architectural specification of how dissolution maps FAI outputs to home mechanism inputs. Different participating Selves may take different things from the same FAI event under their respective home governance authority — asymmetric ingestion is architecturally permitted and does not require symmetric absorption across participants.

4. What is preserved and what is extended

Preserved. Paper 2's three evolution mechanisms operate at home perimeter exactly as Paper 2 §7 specifies. Directed selection requires governance authorization; action-feedback operates through the proposal-and-acceptance machinery with human authority over acceptance; instinct evolution is independent of governance authorization and independent of inter-Self influence. The per-mechanism governance shapes Paper 2 §8 specifies — verification-substrate machinery for instinct integration, authority architecture for DNA evolution, proposal-and-acceptance machinery for action-feedback — carry forward unchanged. Multi-level simultaneous evolution at cell, aspect, and Self scope is unchanged. Horizontal and vertical evolution axes are unchanged. The productive tension between undirected mutation and directed selection is unchanged.

Extended. The source set for action-feedback and directed selection now includes FAI-origin content in addition to home-generated content. The dissolution of a FAI event is an architectural moment at which inter-Self learning enters home evolution machinery under home governance. Asymmetric ingestion is an inter-Self property that has no analog at intra-Self scope: the same FAI event can produce different evolution-feed inputs across participating Selves, depending on what each Self's home governance authorizes. The layer-routing rule at the hand-off boundary — content layer determines which mechanism receives it — is a fresh specification that Paper 2 does not require at intra-Self scope because intra-Self content does not arrive labeled by an external event.

The extension is precisely bounded. Everything else is inheritance.

5. Prior-art inheritance claim

The inheritance claim this note formalizes is stated in its strongest form: no new evolution mechanism is required at inter-Self scope. Any architecture or claim that asserts a novel inter-organizational AI learning mechanism must address that FAI learning enters through Paper 2's established action-feedback and directed selection mechanisms. Both mechanisms are prior art as of Paper 2's publication (April 2026). The four-locus architecture extends them through new input sources; it does not replace or supplement them with new mechanisms.

The protective function of locus 3 is part of this claim. Because instinct evolution is architecturally excluded from inter-Self input, no mechanism exists by which one participating Self's LLM-layer changes propagate to another Self through FAI participation. Inter-Self learning is bounded to the two governed mechanisms — directed selection and action-feedback — both of which require human authorization before they modify a Self's DNA layer. The architecture does not permit ungoverned inter-Self evolutionary influence.

The asymmetric ingestion property is part of this claim. Because each participating Self may absorb different content from the same FAI event under its own home authority, the evolution feed is not a broadcast mechanism that uniformly updates all participants. Each Self evolves through its own governance decisions about what FAI-origin content to absorb. Population-level convergence, if it occurs, emerges from many independent governed decisions across many FAI events — not from a

shared update mechanism operating above the Self level.

The strongest prior-art statement is therefore: Paper 3's four-locus evolution feed is Paper 2's three evolution mechanisms receiving inter-Self inputs. The mechanisms are Paper 2. The four loci are the specification of where and how inter-Self inputs reach those mechanisms. Adversarial claims in either direction — that the mechanisms are novel, or that the loci add nothing — fail on inspection of the architecture's inheritance structure as formalized here.

6. Operational test

A deployment instantiates the $P3 \leftrightarrow P2$ evolution mechanisms inheritance if and only if all of the following hold:

1. FAI-origin action-layer content enters participating Selves' home action layers at dissolution and is processed by the home action-feedback mechanism through the established proposal-and-acceptance machinery — no new machinery is introduced for this processing.
2. FAI-origin DNA-layer content enters participating Selves' directed selection consideration and is absorbed only through governance-authorized acceptance decisions — the same authority architecture that governs intra-Self DNA evolution.
3. No FAI event produces any change in any participating Self's LLM layer or instinct-layer content — instinct evolution remains entirely home and does not receive inter-Self inputs.
4. The routing of FAI outputs to home mechanisms is determined by content layer, not by any mechanism-specific update architecture introduced for inter-Self learning.
5. Each participating Self's absorption of FAI content is independently governed — symmetric absorption across participants is not required and is not the default.

A deployment that introduces additional learning machinery specifically for FAI-derived content, beyond Paper 2's existing evolution mechanisms, is not instantiating this inheritance. A deployment that permits instinct-layer content to cross the inter-Self perimeter is violating locus 3's protective boundary.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

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